Understanding Demographic Drivers of Voter Turnout

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***Abstract*—This analysis examines how demographic characteristics predict voter turnout in the 2020 U.S. general election using a balanced subsample of 2,500 respondents from the 2020 Cooperative Election Study (CES). Employing three modeling approaches—logistic regression with L1 regularization, gradient boosting, and random forest—the analysis finds that demographic variables provide modest but meaningful predictive accuracy (63–67 percent). Across all models, age emerges as the dominant predictor by a substantial margin, with older individuals significantly more likely to vote. Education and income also contribute to turnout, though their effects are smaller and somewhat model dependent. Other demographic factors add limited explanatory value once age, education, and income are considered. Despite these findings, several limitations, including the exclusion of missing data and omission of attitudinal and contextual variables, temper the results. Overall, the results demonstrate that while demographics matter, they are insufficient alone to fully explain voter turnout, highlighting the importance of incorporating broader influences in future research.**

I. INTRODUCTION

Despite decades of research, there is no single consensus model that fully explains voter turnout. Instead, turnout is understood as the result of a complex interaction between a set of factors, including resources, motivation, and social factors [1]. Foundational work in political participation emphasizes that resources, such as education and income, increase individuals' capacity to engage in politics by enhancing civic skills and political knowledge [2]. Subsequent empirical research has reinforced this conclusion, showing that individuals with higher levels of education are significantly more likely to vote across contexts [3], [4].

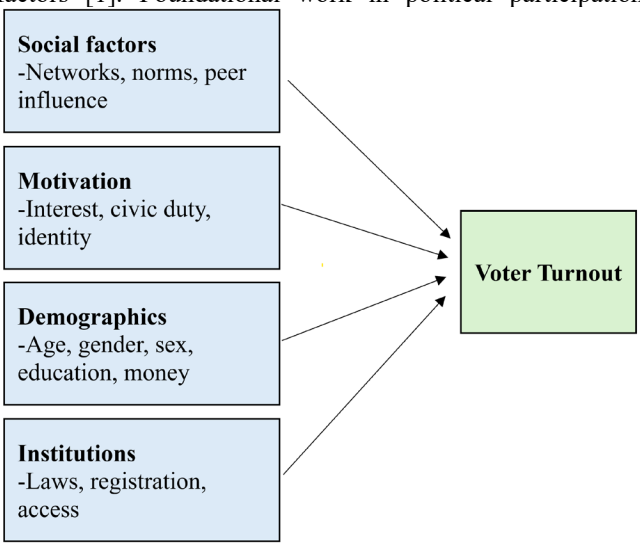

Fig. 1. Conceptual Model of Voter Turnout

The relationship between education and turnout is one of the most extensively studied in literature. Numerous studies demonstrate a strong empirical association between educational attainment and voting behavior, but debates remain about whether this relationship is causal or driven by underlying factors, such as socialization or cognitive ability [3], [5]. More recent research using quasi-experimental designs finds that an additional year of schooling can increase turnout by about three percentage points, providing stronger evidence of a causal effect [6]. At the same time, scholars note that the aggregate relationship between rising education levels and turnout is complex and may depend on relative educational attainment and broader social context [4].

In addition to education, income is a key predictor of turnout, with higher-income individuals consistently voting at higher rates [7]. This relationship contributes to persistent inequalities in political representation, as economically advantaged groups are more likely to have their preferences reflected in electoral outcomes. However, the impact of income is also shaped by contextual factors, such as inequality and mobilization efforts, suggesting that its effects are not purely individual level [7].

Additional demographic variables further explain turnout patterns. Age is among the most robust predictors, with turnout increasing with age as individuals become more politically engaged and socially integrated [1]. Longitudinal studies show that the relationship between education and turnout has also evolved over time, with the effect of higher education becoming stronger in recent decades [8]. Other demographic characteristics, including race, gender, and social integration, also influence participation, often through mechanisms related to access, mobilization, and institutional barriers [1], [2].

Beyond resources and demographics, political engagement plays a central role. Individuals with higher levels of political interest, stronger partisan attachments, and greater exposure to political information are significantly more likely to vote [1]. Importantly, turnout is also shaped by habit formation: individuals who vote in one election are more likely to vote in subsequent elections, reinforcing participation over time [9].

Recent research using large-scale datasets, such as the Cooperative Election Study, further highlights the importance of combining demographic and attitudinal variables in explaining turnout. These studies find that education, income, race, home ownership, and political identity jointly shape voting behavior, with substantial heterogeneity across groups [10].

Modeling voting propensity is critically important for both academic research and practical applications because it provides a systematic way to understand and predict who participates in elections. From a scholarly perspective, predictive models allow researchers to quantify the relative importance of key factors, such as education, income, age, and political engagement, while accounting for complex interactions among them. This helps move beyond descriptive patterns toward more precise, data-driven insights into the mechanisms underlying political participation. From a practical perspective, models of voting propensity are widely used by political campaigns, media, advocacy organizations, and policymakers to identify underrepresented groups, target mobilization efforts, allocate resources efficiently, and predict electoral outcomes. For example, campaigns often use turnout models to prioritize outreach to individuals who are likely persuadable but less likely to vote, thereby maximizing electoral impact. Additionally, understanding voting propensity has important normative implications; by identifying systematic disparities in participation, models can inform efforts to reduce inequalities in democratic representation.

II. METHOD

Because there is no single consensus model that fully explains turnout, this complexity motivates the use of flexible and interpretable modeling approaches that can capture both linear effects and interactions among predictors. This project builds on the existing literature by applying interpretable statistical and machine learning models to the 2020 Cooperative Election Study (CES). By focusing on logistic regression and decision tree models, the analysis aims to identify the most important demographic predictors of turnout while maintaining transparency and interpretability. The demographic predictors that were analyzed are age, educational attainment, family income, race/ethnicity, U.S. Census region, urbanicity, gender, marital status, and home ownership. In doing so, the project seeks to contribute to a deeper understanding of political participation and the persistent inequalities that shape it.

The dataset used for the analysis is the 2020 Cooperative Election Study (CES). The CES is well suited for analyzing voter turnout because it combines large-scale survey data with validated measures of voting behaviors. The CES is one of the largest academic elections surveys in the US, with tens of thousands of respondents, allowing for precise estimation of relationships across diverse demographic groups. A key strength of the CES is its use of validated turnout data, which links survey responses to official voting records where possible, reducing the measurement error and social desirability bias commonly found in self-reported voting [11], [12]. Additionally, the dataset's national scope and representative sampling make it well suited for studying broad patterns in voter participation during the 2020 election.

The 2020 CES dataset contains data on 61,000 respondents, with fewer than 15 percent missing responses for any of the selected demographic predictors, indicating that these variables are mostly complete. Accordingly, incomplete cases were dropped. From the remaining complete cases, a random sample of 2,500 respondents was drawn, stratified to include an equal number of individuals who did and did not vote in the 2020 general election. This subsampling reduces computational demands, allowing for the creation of more models and more complex specifications, such as deeper trees and additional boosting iterations, with the same limited resources. Categorical demographic variables with very small group counts were consolidated because low counts can cause overfitting, high variance in logistic regression coefficients, and reduced interpretability. Dimensionality reduction techniques were not applied to preserve interpretability, particularly given that the total number of predictors is moderate and that logistic regression models with L1 regularization and decision tree models can handle numerous predictors while selecting the most informative ones. Finally, predictors were scaled prior to estimating logistic regression models to ensure comparability of units, allowing coefficients to be directly interpreted in terms of their relative influence on the log odds of voting.

III. RESULTS

The 2020 CES dataset contains 61,000 respondents, of whom 64 percent voted in the 2020 general election. To facilitate balanced classification, a subsample of 2,500 respondents was constructed with an equal split between voters and nonvoters. Exploratory bivariate analysis reveals several clear relationships between demographic characteristics and voter turnout. Most noticeably, age, education, and income are all positively associated with voting, which is consistent with previous research. In the sample, the median age of voters is 55 years old, compared to only 38 years old for nonvoters, indicating a substantial age gap. This suggests that nationally, older individuals are significantly more likely to participate in elections than younger individuals.

A similar pattern emerges for education. Among respondents who voted, 44 percent have a four-year college degree, compared to only 26 percent of nonvoters. This large difference reinforces the well-documented relationship between higher levels of education and increased political participation.

Income also exhibits a positive relationship with turnout. The median family income among voters is $65,000, compared to $45,000 for nonvoters. This gap suggests that nationally, higher-income individuals are more likely to vote.

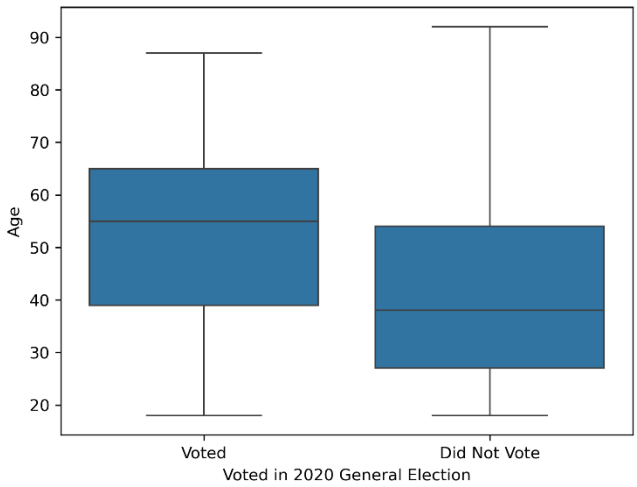

Fig. 2. Boxplots of Age by 2020 Voting Status (Sample Respondents)

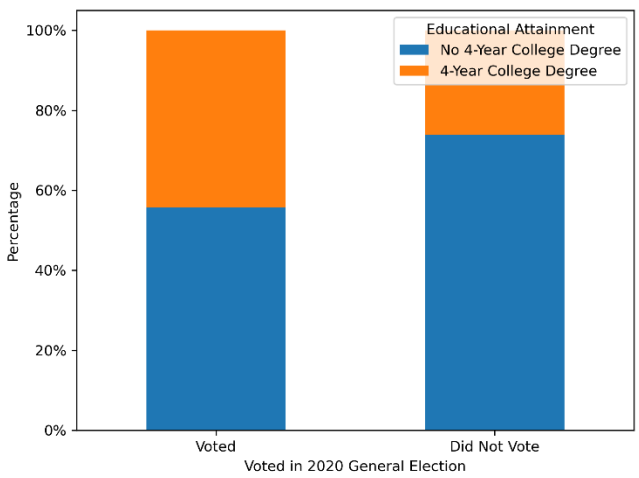

Fig. 3. Bar Charts of Educational Attainment by 2020 Voting Status (Sample Respondents)

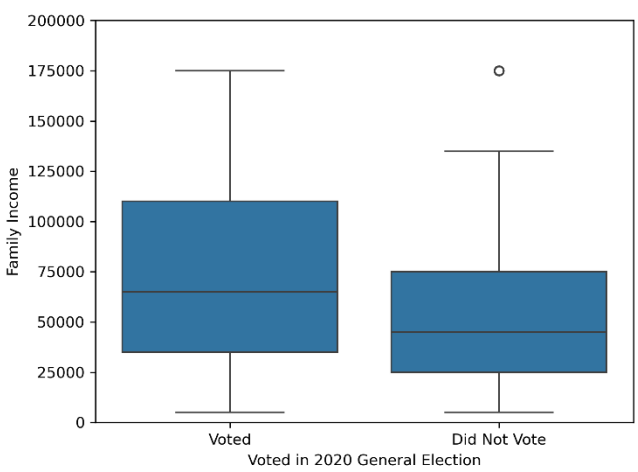

Fig. 4. Boxplots of Family Income by 2020 Voting Status (Y-axis truncated to $0–$200k) (Sample Respondents)

While these bivariate relationships are informative, they do not isolate the independent effect of each variable. Thus, multivariate models were created to account for the joint influence of multiple predictors. This is important because observed relationships may be partially or fully explained by correlated variables. For example, the observed relationship between voting and income may be explained solely by voters having higher education than nonvoters and individuals with higher education making more.

Three classification models were implemented—a gradient boosting model, a random forest model, and a logistic regression model with L1 regularization. Each model used nine demographic predictors: age, educational attainment, family income, race/ethnicity, U.S. Census region, urbanicity, gender, marital status, and home ownership. All three models achieved testing accuracies between 63 and 67 percent, indicating meaningful predictive power relative to random guessing. However, all models showed some level of overfitting, as seen by higher test accuracy than training accuracy. The random forest model, in particular, achieves near-perfect accuracy on the training data (99.35 percent) but performs substantially worse on the test data (63 percent), suggesting that it captures noise in the training set rather than generalizable patterns. This is consistent with known tendencies of random forests to overfit when not sufficiently regularized.

Across all three models, age emerges as the most influential predictor of voting behavior by a considerable margin. In the logistic regression model, age had the largest magnitude coefficient (0.636), indicating that a one standard deviation increase in age increases the log-odds of voting by 0.636, holding all other demographic variables constant. This suggests a large effect as the odds of voting increase by almost 90 percent as an individual's age increases by one standard deviation (17.40 years in the training dataset), all else equal. Similarly, age has the highest feature importance in both the gradient boosting (0.591) and random forest (0.375) models, indicating that age contributes strongly to reducing prediction error. In the gradient boosting model, age alone accounts for over half of the total reduction in prediction error, underscoring its dominant role in explaining turnout. The next highest importance was income at 0.216 for random forest and education at 0.141 for gradient boosting, which are significantly below the importance of age. This indicates that not only is age the most influential predictor in predicting turnout, it is the most influential by a long shot.

TABLE I
MODEL PERFORMANCE COMPARISON

Predictors	**Coefficient/Feature Importance**		
	Logistic regression	***Gradient boosting***	***Random forest***
Age	0.636 [1]	0.591 [1]	0.375 [1]
Education	0.352 [2]	0.141 [2]	0.046
Income	0.103	0.137	0.216 [2]
Home ownership	-0.141	0.043	0.043
Gender	-0.042	0.035	0.047
Race/ethnicity (Black, Hispanic, Other)	[-0.036, -0.105, -0.093]	0.025	0.067
U.S. Census region (Midwest, South, West)	[0.047, -0.055, 0]	0.018	0.086
Urbanicity (Suburb, Town, Rural)	[0, 0, 0]	0.008	0.083
Marital status	0	0.002	0.036

After age, education and income are the next most influential predictors of voting, although their relative importance varies across models. In the logistic regression model, both variables are positively associated with voting, with education (0.352) having a stronger effect than income (0.103). Practically, having a four-year college degree is associated with a 42 percent increase in the odds of voting, all else equal. Also, a one standard deviation ($67,188 in the training dataset) increase in an individual's income increases the odds of voting by 10 percent, all else equal. Home ownership shows a slightly stronger effect than income (-0.141), but it has low feature importance in both gradient boosting and random forest (both 0.043). This difference reflects how the models define importance. The L1-regularized logistic regression captures the feature's linear association with voting, while tree-based models assess how much a feature improves predictive performance through splits. Consequently, features with smooth, linear effects or high correlation with other predictors may receive relatively large coefficients in logistic regression but low importance in gradient boosting and random forest.

The gradient boosting model shows similar importance for education (0.141) and income (0.137). This shows that both variables are similarly important predictors of voting, but education is slightly more influential. However, the random forest model assigns much greater importance to income (0.216) than education (0.046), indicating a divergence in how these models capture underlying relationships. This discrepancy likely reflects differences in model structure. Gradient boosting builds trees sequentially, allowing it to capture incremental and additive effects across many weak predictors. As a result, a binary variable, like education, may have a consistent but relatively modest effect on voting that is accumulated across many trees, making it appear more important in the boosting model. In contrast, random forests rely on averaging independently constructed trees and tend to prioritize variables that produce strong, early splits. Thus, income, measured continuously, may create larger reductions in impurity early in the trees, leading to higher measured importance, while the simpler structure of education makes it less frequently selected. This suggests that income may have a stronger or more direct relationship with voting, while education's influence may be more nuanced, potentially operating through smaller or more context-dependent effects that are better captured by boosting. However, these differences highlight that feature importance measures are model-dependent and may be biased toward variables with greater variability or more potential split points. Thus, these results should be interpreted cautiously.

Beyond these key predictors, several variables exhibit relatively low importance across models. In the L1-regularized logistic regression model, five coefficients (three indicators for urbanicity, west compared to northeast U.S., and married compared to unmarried) were shrunk to 0, indicating that they do not contribute additional predictive power beyond other included variables under the regularization constraint. However, this does not imply that these variables are unrelated to turnout, but rather, that their predictive information may be weak, indirect, or captured through correlated predictors. The tree-based models corroborate this interpretation, as these variables receive low feature importance scores.

Overall, the results consistently demonstrate that age is the dominant predictor of voting behavior, with education and income playing secondary but still meaningful roles. Other demographic variables contribute relatively little to predictive accuracy once these primary factors are accounted for. These results matter because they clarify whether explanatory and predictive power lie in models of voting behavior, allowing researchers and practitioners to focus attention and resources more efficiently. If age consistently dominates, it suggests that life-cycle effects, such as political socialization, stability, and stake in long-term policy outcomes, are more influential than many commonly cited demographic factors. The secondary roles of education and income still point to meaningful stratification in participation, but their diminished importance relative to age indicates that interventions aimed at increasing turnout may be most effective when tailored to different age groups rather than broadly targeting socioeconomic status alone. At scale, this insight can improve the design of voter outreach strategies, survey weighting schemes, and predictive models by placing greater emphasis on age as a key organizing factor, leading to more accurate forecasts and more targeted civic engagement efforts without overcomplicating models with variables that add limited marginal value.

IV. CONCLUSION

This analysis aimed to identify which demographic characteristics most strongly predict voter turnout in the 2020 general election. Using a balanced subsample of 2,500 respondents drawn from the CES, both exploratory and multivariate analyses provide consistent evidence that demographic characteristics meaningfully structure voting behavior. Across all three modeling approaches—logistic regression with L1 regularization, gradient boosting, and random forest—predictive performance was modest but clearly above chance, with test accuracies ranging from 63 to 67 percent. These results indicate that demographic variables alone contain meaningful, though incomplete, information about turnout.

Substantively, the results demonstrate that age is the dominant predictor of voting behavior by a wide margin. In the logistic regression model, a one standard deviation increase in age is associated with an almost 90 percent increase in the odds of voting, holding all other demographic predictors constant. This finding is reinforced by the tree-based models, where age accounts for the largest share of feature importance (0.591 in gradient boosting and 0.375 in random forest), far exceeding any other variable. Education and income emerge as secondary predictors. Having a four-year college degree is associated with a 42 percent increase in the odds of voting, while a one standard deviation increase

in an individual's income corresponds to a 10 percent increase in the odds of voting, all else equal. Feature importance measures from the gradient boosting model suggest that these two variables contribute similarly to predictive performance (0.141 and 0.137, respectively). However, the random forest model places substantially greater weight on income (0.216) than education (0.046), underscoring the model-dependent nature of such metrics. Other variables, including gender, race/ethnicity, urbanicity, marital status, and homeownership, contribute relatively little additional predictive power once age, education, and income are taken into account.

Several limitations qualify these findings. First, the use of a balanced subsample, while useful for classification, means that the results do not reflect the true population turnout rate and may affect model calibration. The same is true for the overall 2020 CES data, as minority populations are oversampled. Second, the analysis is restricted to demographic variables and therefore omits important attitudinal, institutional, and contextual factors that are known to influence turnout. Third, the analysis excludes all respondents with missing values on any demographic variable, which may introduce selection bias if missingness is systematic rather than random. Future research could use techniques to fill in potential values for this missing data and see if the results change. Fourth, evidence of overfitting suggests that some results may not generalize as well out of sample. Fifth, the dataset only includes data on the 2020 general election and may not generalize to other election cycles. Sixth, feature importance measures are inherently model-specific and can be biased by variable scaling and measurement, limiting direct comparability across models. Finally, because the analysis is observational, the results should be interpreted as associational rather than causal.

Taken together, the findings directly address the analysis's objective by demonstrating that while multiple demographic factors are associated with turnout, age overwhelmingly accounts for the largest share of explained variation, with education and income playing important but secondary roles. At the same time, the moderate predictive performance of all models underscores that turnout is not determined by demographics alone. This suggests that future research seeking to better explain and predict voting behavior should integrate demographic characteristics with attitudinal and contextual variables to capture the full complexity of political participation.

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